Shaky structures: The wobbly world of causal graphs in software analytics

An Empirical Audit of 23 Datasets and 4 Causal Graph Generators

Jeremy Hulse (NCSU),
Nasir U. Eisty (U.Tennessee)
Tim Menzies (NCSU)

prof, cs, **ncstate**, usa
`http://timm.fyi`

Apr 2025

Motivation: The Allure of the "Why"

Claim: Software Engineering (SE) is tired of correlations; we want **causes**.

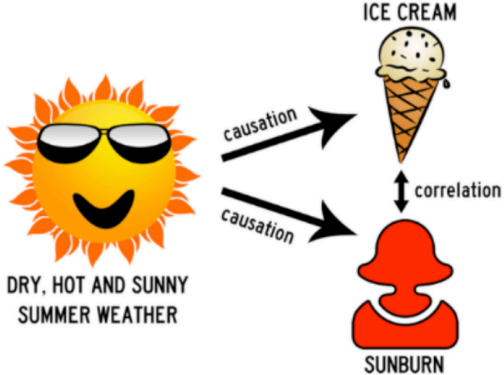
- We don't want to know that "Size correlates with Defects."
- We want to know: "If I change this module, will I stop the bugs?"
- Causal Discovery Algorithms (CDAs) promise to find these hidden structures automatically.

But as the "just throw an LLM at it" phase ends, so must the "just throw a CDA at it" phase. [cite: 3]

Ronald Fisher:

Correlation is not Causation

(but can we access "causation"?)



Cigaret Cancer Link Is Bunk

By CHARLES GOLDEN

A grim warning has been stamped on every pack of cigarettes sold in the United States since Jan. 1, 1966: "Caution: Cigarette Smoking May Be Hazardous to Your Health."

But the truth is that 50 years of intensive research has failed to produce evidence that smoking causes cancer or any other disease.

If stone cold facts have not prompted the government into ending this regulation, had their warning been accepted by the U.S. Public Health Service instead, every brand of cigarette would have been outlawed by the U.S. Public Health Service long ago. Yet the strict propaganda warning of the Surgeon General that the health agency promulgates, which has been based on a belief that cigarettes actually may cause the great causes of lung cancer and heart disease, may be false.

Medical studies have indicated that cigarettes actually may have been a boon to smokers' health.

Medical studies have indicated that cigarettes actually may have been a boon to smokers' health.

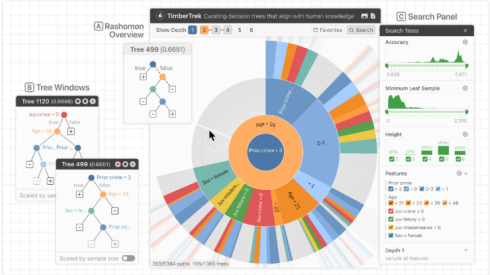
Medical studies have indicated that cigarettes actually may have been a boon to smokers' health.

70 Million Smokers Have Been Victims Of a False Alarm

FALSE ALARM: Leading medical men say smoking is not true.

Rashomon Volumes

Given so many good models, epistemically dishonest to reason about "the" model



(a) Rashomon set of the 1,365 best sparse decision trees for the COMPAS dataset, generated by TreeFARMS and TimberTrek (figure from Wang et al. 20

Amazing Things Come From Having Many Good Models

Cynthia Rudin [1] * Chudi Zhong [1] Lesia Semenova [1] Margo Seltzer [2] Ronald Parr [1] Jiachang Liu [1] Srikar Katta [1] Jon Donnelly [1] Harry Chen [1] Zachery Boner [1]

Abstract

The *Rashomon Effect*, coined by Leo Breiman, describes the phenomenon that there exist many equally good predictive models for the same dataset. This phenomenon happens for many real datasets and when it does, it sparks both magic and consternation, but mostly magic. In light of the Rashomon Effect, this perspective piece proposes reshaping the way we think about machine learning, particularly for tabular data problems in the nondeterministic (noisy) setting. We address how the Rashomon Effect impacts (1) the existence of simple-yet-accurate models, (2) flexibility to address user preferences, such as fairness and monotonicity, without losing pre... (3) uncertainty in prediction... nations, (4) relat... gorith...

think of the Rashomon Ef... from getting a...

The Rashomon Effect for Visualizing High-Dimensional Data

Yiyang Sun* Haiyang Huang*+ Gaurav Rajesh Parikh* Cynthia Rudin

Efficient Rashomon Set Approximation for D... Trees

Zakk Heile
Department of Computer Science
Duke University
Durham, NC 27708
zakk.heile@duke.edu

Hayden McTavish
Department of Computer Science
Duke University
Durham, NC 27708
hayden.mctavish@duke.edu

Varun Babbar
Department of Compute...
Duke Universit...
Durham, NC 27...
varun.babbar@du...

Cynthia Ru...
Department of Com...
Duke Univ...
Durham, NC...
cynthia@cs...

The Reality: The Instability Problem

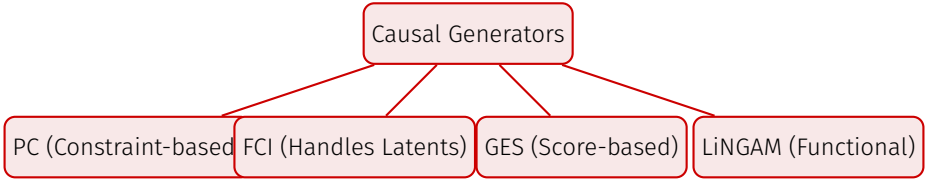
Claim: Automatic causal graphs are currently too unstable to trust for industrial SE.

The Definition of Instability

Small changes to the input data result in **large, unpredictable changes** in the resulting graph.

- If the graph changes every time you re-run it, it isn't an "explanation."
- It is just noise masquerading as a diagram.

We audited the four most prominent paradigms in causal discovery. They all share the same "Unfair Advantage" of automation—and the same flaws in stability. [cite: 7]



The Archive: 23 Datasets across 30 Years

- We didn't test on toy data. We used the **PROMISE** and **JURECZKO** repositories.
- **Breadth:** Ant, Camel, JEdit, Lucene, POI, Tomcat, Velocity, and more.
- **Depth:** Multiple releases per project (e.g., Ant 1.3, 1.4, 1.5, 1.6, 1.7).

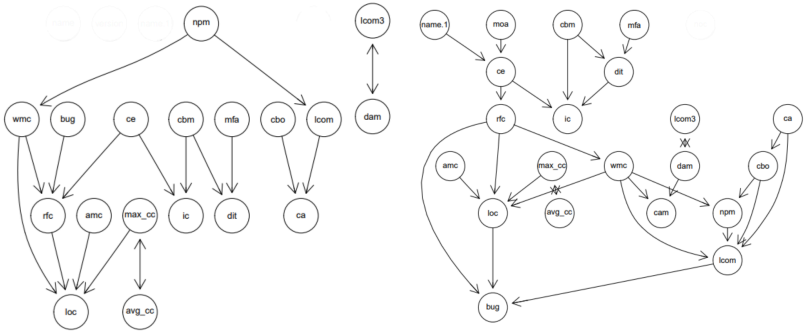
Not a green-field project

This is a audit of how these tools behave on real-world SE evolution. [cite: 11]

The Yardstick: The Jaccard Index

- $J(A, B) = \frac{|E_A \cap E_B|}{|E_A \cup E_B|}$
- $J = 1.0$: The graphs are identical.
- $J = 0.0$: The graphs have zero edges in common.
- **Rule:** If J is low, your "Why Engine" is broken. [cite: 25]

We measured Jaccard across versions, across tools, across data subsets. e.g. v5,v7:



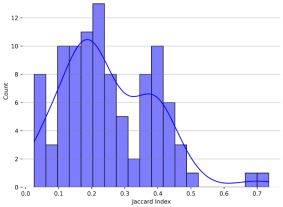
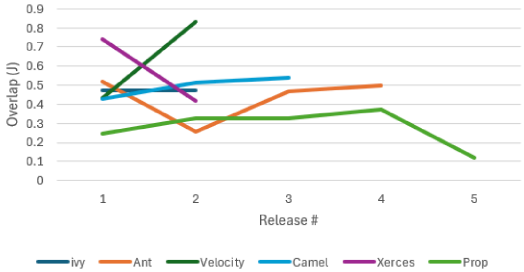


Fig. 2: Jaccard index results seen in 100 experiments where two causal graph generators are applied to the same data set.

Finding 1: The Time Flip

Question: Does the "cause" of defects stay the same from Version 1 to Version 2?

- **The Result:** In nearly all cases, over half the edges vanish between releases.
- Median stability across versions: **Extremely Low**.
- **The Risk:** A manager fixes a "cause" in Version 1.5 that no longer exists in Version 1.7.



Finding 2: The Domain Gap

Question: If I learn the causal graph for Camel, can I use it for Tomcat?

- **The Result:** Over two-thirds of causal connections differ between projects.
- This suggests causal discovery in SE is **highly idiosyncratic**.
- Generalization is a dangerous assumption in this landscape.

Table 6: **RQ2 results:** Inter-Project Jaccard Values. This matrix is symmetrical across the diagonal (so the values of blank cell i, j can be found in cell j, i).

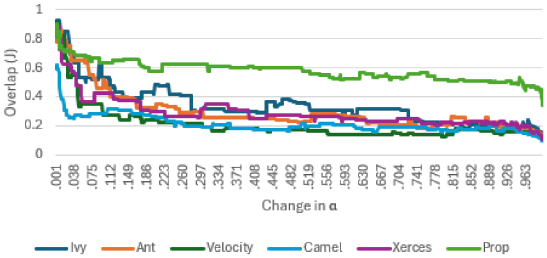
Project	**Ivy**	**Ant**	**Velocity**	**Camel**	**Xerces**
Ivy	1	.28	.37	.25	.26
Ant		1	.28	.19	.20
Velocity			1	.20	.26
Camel				1	.29
Xerces					1

Finding 3: The Fragility of Alpha

Question: Does changing the significance level α (e.g., 0.01 to 0.05) break the graph?

- **The Result:** Minor tuning adjustments generate radically different causal structures.
- These tools are **not robust** to standard hyperparameter variations.

Fig. 4: **RQ3 results:** Graph of Jaccard values when shifting alpha $\Delta = 0.001$. All graphs compared to the graph from $\alpha = 0.001$.



Key Design Warning

If your "Why" depends on whether $\alpha = 0.01$ or 0.05, you don't have a "Why." [cite: 25]

Finding 5: Other Data? Other Causal Generators?

$$\underbrace{\begin{pmatrix} RQ1 \\ RQ2 \\ RQ3 \\ RQ4 \end{pmatrix}}_{4\ \textit{research questions}} \times \underbrace{\begin{pmatrix} PC \\ FCI \\ GES \\ LiNGAM \end{pmatrix}}_{4\ \textit{generators}} \times \underbrace{\begin{pmatrix} 18\ \textit{defect prediction} \\ 2\ \textit{software configuration} \\ 3\ \textit{software process management} \end{pmatrix}}_{3\ \textit{tasks},\ 23\ \textit{data sets}}$$

Table 7: Jaccard indexes seen in different experiments.

		Causal graph generator			
data	**question**	**PC** μ (σ)	**FCI** μ (σ)	**GES** μ (σ)	**LiNGAM** μ (σ)
defect	**RQ1** : across releases	.67 (.07)	.43 (.11)	.50 (.08)	.39 (.06)
defect	**RQ2** : across projects	.42 (.04)	.19 (.07)	.34 (.04)	.24 (0.5)
defect	**RQ3** : changing α	.51 (.06)	.27 (.07)	N/A	N/A
config	**RQ3** : changing α	.27 (.07)	.10 (.04)	N/A	N/A
process	**RQ3** : changing α	.12 (.03)	.06 (.02)	N/A	N/A
defect	**RQ4** : subsets	.84 (.06)	.59 (.12)	.70 (.07)	.52 (.10)
config	**RQ4** : subsets	.75 (.18)	.59 (.26)	.73 (.16)	.57 (.20)
process	**RQ4** : subsets	.64 (.11)	.54 (.13)	.59 (.09)	.29 (.07)

The Why: Why Do Graphs Fail in SE?

- **Data Sparsity:** SE data is often "fat and short" (many features, few rows). [cite: 9]
- **Noisy Metrics:** Static code features are proxies, not direct physical laws.
- **Stochastic Nature:** Software evolution is driven by human choice, not just mathematical necessity. [cite: 37]

We need **hybrid workflows** that combine these graphs with symbolic checks. [cite: 5, 25]

Roadmap: Moving Beyond the "Hype"

1. **Intersection Filters:** Only report edges found by *at least two* different generators.
2. **Stability Audits:** Always run a "10% drop test" before publishing a graph.
3. **Type-Safe Checks:** Use Z3 to reject causal links that violate known SE constraints. [cite: 14, 25]
4. **Ensemble Causality:** Treat causal discovery as an optimization problem over an ensemble of graphs. [cite: 33]

Why Me, Why Now

- We have the archive: 23 projects over 30 years. [cite: 35]
- We have the machinery: thousands of GPU hours auditing these generators. [cite: 36]
- **The Conclusion:** Causal graphs are a powerful *hypothesis generator*, but a dangerous *decision maker*—until we solve the instability problem.

Closing pitch

This is the moment to connect causal mining with symbolic safety and stochastic rigor. [cite: 38]